

Ten Layers at the Counterloop/Quotient Frontier

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Abstract

RH-272–RH-281 reorganize the post-RH-271 route around an exact deterministic monodromy counterloop and a separate noisy spectral quotient. The counterloop branch has an all-order fixed-coefficient bridge, a sharp minimal rank theorem, and a precise aggregate Fourier transport requirement. The archived seven-row cloud audit does not certify that requirement. On the operator side, the raw Hilbert–Schmidt mass diverges like $\sigma^{-1/2}$, fixed-rank zero-noise quotients cannot contract in the Calkin algebra, local positive-noise shell charts do work, and a variable-rank block-power criterion is available. The spectral vector remains $(0, 0, 0, 1, 1)$; the separate counterloop vector is $(1, 1, 0, 1, 1)$. Both complete counts are zero and Gates A–E remain false/open.

1 The ten layers

paper	result	first missing input
RH-272	exact monodromy counterloop bridge	noisy spectral identification
RH-273	minimal rank and unique equality shell	aggregate transport
RH-274	Fourier criterion; max-phase firewall	actual low-frequency control
RH-275	seven-row floating audit	interval/asymptotic certificate
RH-276	sharp raw $\mathcal{S}_2$ mass law	cancellation or rank growth
RH-277	fixed-rank Calkin no-go	rank-growing quotient
RH-278	local positive-noise shell activation	zero-noise gluing
RH-279	variable-rank block-tail theorem	uniform block data
RH-280	dual ledger	one branch must gain its tail
RH-281	scoped synthesis	new operator input

2 Logical separation

The exact counterloop is a graded atomic superloop. RH-242 permits this object without a projector, but an ordinary determinant quotient requires the atoms to be an actual spectral submultiset of the noisy operator. The latter identification is absent. Conversely, RH-278 supplies local positive-noise spectral charts without identifying the deterministic counterloop. These are different branches and must not be merged by finite matching.

3 Operator frontier

RH-276 proves $\sigma\|A_\sigma\|_2^2 \rightarrow (2\sqrt{\pi}r_H^2)^{-1}$, and RH-277 proves that finite-rank compression of the deterministic Koopman isometry has essential power norm one (larger than one after Hardy scaling). Therefore the RH-269 fixed-rank zero-noise package is a closed route in this natural stationary L^2 geometry. RH-279 gives the correct replacement: a moving block length and direct trace-norm power bounds.

4 Ledger and gates

$$v_{\text{spectral}} = (0, 0, 0, 1, 1), \quad v_{\text{counterloop}} = (1, 1, 0, 1, 1).$$

Neither vector is complete. No theorem in this batch constructs a Hilbert–Polya operator, identifies a Riemann zero, proves a von Mangoldt trace formula or completed-zeta divisor equality, or implies RH. Gates A–E remain false/open.